

CHEUNG Hiu Ching, Athena

Robotics Researcher

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EDUCATION

• The Hong Kong Polytechnic University (HKPolyU)

2022-2025

Master of Philosophy (Aeronautical and Aviation Engineering)

- Thesis title: Design and Control of a Soft Aerial Vehicle for Conducting Aerial Grasping
- Supervised by Prof. Chih-yung WEN (AAE) and co-supervised by Dr Henry K. CHU (ME)
- Affiliated with the High-speed Thermo-fluid and MAV/UAV Lab (AIRo Lab) at the Research Centre for Unmanned Autonomous Systems (RCUAS)

• The Hong Kong Polytechnic University

2018-2022

Bachelor of Engineering (Honours) in Mechanical Engineering

- Dean list (2019/2020)
- FYP title: Development of an Aerial Air Quality Monitoring Platform Based on Vertical Takeoff and Landing (VTOL) Unmanned Aerial Vehicle (UAV) (Supervised by Prof. Chih-yung WEN)
- Virtual summer exchange: Girton College, Cambridge - Mathematics for Engineering Online Summer Programme, 2021

PUBLICATION

- **H. C. Cheung**, W. Yue, Z. Han, M. Chen, G. Cao, H. Liu, H. Ren “Learning-based Hierarchical Tracheal Anatomy Understanding from Sparse Surgical Demonstration Annotations for Ultrasound Robots,” *International Conference on Cyborg and Bionic Systems (ICCBS)* 2026, Macau, China, 2026, doi: 10.1088/1742-6596/3291/1/012035.
- W. Yue[†], R. Zhang[†], K. Sun[†], **H. C. Cheung**, R. Zhao, Y. Zhang, C. Xu, W. Yuan, H. Ren “Bio-inspired Self-Oscillating Needle Transducer with Unified In-Keyhole Sensing and Actuation Capability,” (*Accepted by IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* 2026), 2026.
- Y. Zhang[†], W. Yue[†], **H. C. Cheung**, H. Meng, X. Yang, S. Fu, J. Su, H. Gao, C. Chen, X. Zuo, H. Ren “Bilateral Soft Pouch Matrix Actuation for Multimodal Intraluminal Locomotion of Liquid Sampling Robotics,” *IEEE Transactions on Medical Robotics and Bionics*, 2026, doi: 10.1109/TMRB.2026.3699220.
- D. Yulianti[†], R. Prakash[†], **H. C. Cheung**, A. K. Strong, P. J. Codd, S. Lin, “PalpAid: Multimodal Pneumatic Tactile Sensor for Tissue Palpation,” *2026 IEEE 9th International Conference on Soft Robotics (RoboSoft)*, Kanazawa, Japan, 2026, pp. 660-665, doi: 10.1109/RoboSoft67810.2026.11522855.
- **H. C. Cheung**, “Design and control of a soft aerial vehicle for conducting aerial grasping,” *The Hong Kong Polytechnic University*, 2025. (MPhil Thesis)
- **H. C. Cheung**, B. Jiang, Y. Hu, H. K. Chu, C.-Y. Wen, and C.-W. Chang, “Aerial grasping with soft aerial vehicle using disturbance observer-based model predictive control,” *arXiv preprint arXiv:2409.14115*, 2024.
- **H. C. Cheung**, C.-W. Chang, B. Jiang, C.-Y. Wen, and H. K. Chu, “A modular pneumatic soft gripper design for aerial grasping and landing,” *2024 IEEE 7th International Conference on Soft Robotics (RoboSoft)*, San Diego, CA, USA, 2024, pp. 82-88, doi: 10.1109/RoboSoft60065.2024.10521918.
- C.-W. Chang, L.-Y. Lo, **H. C. Cheung**, Y. Feng, A.-S. Yang, C.-Y. Wen, and W. Zhou, “Proactive guidance for accurate uav landing on a dynamic platform: A visual-inertial approach,” *Sensors*, vol. 22, no. 1, p. 404, 2022.

AWARDS AND SCHOLARSHIP

• HKSAR Government Scholarship Fund - Endeavour Merit Award

2018/19 - 2023/24

• The Hong Kong Jockey Club Scholarships –Undergraduate Scholarship

- The Hong Kong Jockey Club

2019/20 - 2021/22

• BEA Inspiring Student Scholarship

- Bank of East Asia

2019/20

- **Best Engineering Design Award** The Robocon 2019 Hong Kong Contest
- Hong Kong Science and Technology Parks Corporation 2019
- **HKSAR Government Scholarship Fund - Talent Development Scholarship** 2018/19 -2019/20
- **Hong Kong Top 10 Outstanding Teens Award** Hong Kong Outstanding Teens Election
- Hong Kong Playground Association and The Outstanding Young Persons' Association 2016

WORK EXPERIENCE

- **The Chinese University of Hong Kong** Sep 2025 - Aug 2026
Junior Research Assistant (Full-time) / Supervisor: Prof. Hongliang REN Department of Electronic Engineering
– Affiliated with the Lab for Medical Robotics, Embodied AI, Navigation in vivo, Intelligent Medical Mechatronics (REN Lab)
– Designed pneumatic systems and tactile sensing modules for the palpation project
– Developed learning-based hierarchical tracheal anatomy understanding from sparse surgical demonstration annotations for ultrasound Robots
- **The Hong Kong Polytechnic University** Jan 2025 - Aug 2025
Research Administrative Assistant (Full-time) / Supervisor: Prof. Chih-yung WEN AIRO Laboratory
– Associating with the project "Research Centre for Low Altitude Economy" (4-CE0Q)
– Assisting with a book ("New Space: From Low Earth Orbit to the Moon and Beyond") that is to be published, addressing tasks such as formatting the citations and reference lists, modifying the diagrams
- **The Hong Kong Polytechnic University** May 2023 - Aug 2024
Project Technical Assistant (Part-time) / Supervisor: Prof. Chih-yung WEN AIRO Laboratory
– Had associated with the research project "Research Centre for Unmanned Autonomous Systems" (P0046487)
– Provided technical support for 3D printing and mechatronics design
- **Hong Kong Center for Construction Robotics** Jan 2023 - Jun 2023
Research Assistant (Part-time) / Supervisor: Dr Ching-Wei Chang
– Provided technical support for 3D printing
– Designed the mechanical structure of products and drew the 3D CAD drawings
- **The Hong Kong Polytechnic University** Sep 2022 - May 2023
Project Assistant (Part-time) / Supervisor: Prof. Chih-yung WEN AIRO Laboratory
– Had associated with the research project "Research Centre for Unmanned Autonomous Systems" (P0046487)
– Provided technical support for 3D printing and composite manufacturing (Carbon fiber airframes)
- **Hong Kong Center for Construction Robotics** Jun 2022 - Aug 2022
Student Helper (Full-time) / Supervisor: Dr Ching-Wei Chang
– Joined one of the existing start-up teams, which is focusing on construction robots
– Designed the mechanical structure of products and drew the 3D CAD drawings
- **The Hong Kong Polytechnic University** Aug 2021 - May 2022
Student Assistant (Part-time) / Supervisor: Prof. Chih-yung WEN AIRO Laboratory
– Had associated with the research project "Trial: Development of Vertical Take-Off and Landing (VTOL) Unmanned Aerial Vehicle (UAV) for Air Quality Monitoring in Greater Bay Area" (K-ZPJU)
– Provided technical support for 3D printing
– Designed the mechanical structure of a movable landing platform for UAVs and controlled its movement with Arduino programming
- **The Hong Kong Polytechnic University** Dec 2020 - Jul 2021
Student Assistant (Part-time and Full-time) / Supervisor: Dr Henry Kar Hang CHU Biomimetic Robotics Laboratory
– Automated pick-and-place task with object recognition using deep learning
– Applied vision-based control for a robot arm (UR5) and conducted system calibration to ensure precise control

- Incorporated deep learning techniques, specifically Convolutional Neural Networks (CNN), for grasping random objects

• **Carmel Divine Grace Foundation Secondary School**

Sep 2018 - Jun 2020

Robotics Team Coach (Part-time)

Hong Kong

- Led students to win in the International Robotic Olympiad 2019
- Taught students how to use CAD (Computer Aided Drawing) (2D: CorelDRAW and 3D: SolidWorks)
- Taught students how to build robots with DC gear motors, servo motors, and micro-controllers

VOLUNTEER SERVICES AND EXTRA-CURRICULAR ACTIVITIES

• **Researcher, ASU MARGIN Lab**

Mar 2025 - now

- Served as a student organizer for the "2nd International Workshop on the Evolving Landscape of Surgical Robotics (ELSR 2026): Surgical Autonomy for Today and Tomorrow" at IROS 2026 (<https://sites.google.com/view/iros-elsr-2026>)
- Contributing to lab projects focused on soft robotics, mechatronics, and robot manipulation
- Maintaining the lab website

• **Judge and Organizer, Hong Kong Robotics Club**

Apr 2018 - now

- Demonstrating judgment in the Hong Kong Robotic Olympiad and International Robotic Olympiad (<https://robotorghk.weebly.com/>)
- Tutoring in InnoTech Workshops in InnoCarnival (2013 to 2025)
- Contributing to the game rules modification, preparation of the competitions, and event follow-ups

• **Wooden Boat Crafting Skills - Intern, The Warehouse Teenage Club**

Jun 2024 - Jun 2025

- Intangible Cultural Heritage Documentation and Promotion Project
- Engaged in learning and participating in the process of wooden boat crafting to ensure compliance with relevant safety standards and requirements.
- Involved in documenting the craft of wooden boat making, meticulously recording the production process and various details.

• **Internal Vice President, Outstanding Teens Association (Hong Kong)**

Oct 2020 - Aug 2024

- Contacted and promoting events to OTA members
- Handled Financial management
- Coordinated internal administration and organized external voluntary services (e.g., University Simulations in 2021 (<https://skmdonaldshek.wixsite.com/hkotausims2021>) and 2022 (<https://skmdonaldshek.wixsite.com/hkotausims2022>))

• **Team member, HKPolyU Unmanned Aerial Vehicles Team**

Sep 2019 - Aug 2022

- Had been prepared for UAV Challenge –Medical Rescue:
 - A mechanical structure was designed to release a ground vehicle from a fixed-wing VTOL
 - Developed a geofence system design for the fixed-wing VTOL
- Built a fixed-wing VTOL (vertical take-off and landing) unmanned aerial vehicle with postgraduate teammates (Mini Talon: <https://youtu.be/ELSqvWizsCc>, start from 00:57-01:50)

• **Vice President (Executive), Outstanding Teens Association (Hong Kong)**

Aug 2018 - Sep 2020

- Led the team of the Academic and Development Committee to plan and organize a Peer to Peer Programme (P2P X STEM) for all students in primary schools and secondary schools in Hong Kong
- Contacting with other student leaders and teachers to promote P2P X STEM
- Sharing personal experience on STEM (robotics) in P2P X STEM

• **Leader of the Team, Crimson, HKPolyU FENG Robotics Club**

Oct 2018 - Oct 2019

- Chief designer of Manual Robot 1 (A multi-tasking robot with several types of actuators)
- Designer of the rack of compressed air tank for pneumatic cylinders
- Won Best Engineering Award in the Robocon 2019 Hong Kong Contest

TECHNICAL SKILLS AND INTERESTS

Languages: English, Cantonese, Mandarin

CAD & CFD: AutoCAD, CorelDRAW, Fusion 360, SOLIDWORKS, TinkerCAD, Ansys Fluent

Programming language libraries & Frameworks: Arduino, C++, Python, ROS, OpenCV, ArduPilot, PX4, TensorFlow, micro:bit, MIT App Inventor, LEGO EV3, Tello TT

MEDIA

• Interview Video, Student Affairs Office @ PolyU

Jun 2022

SEN Students' Sharing

– A Big Fan of Robotics

[https://www.polyu.edu.hk/sao/student-resources-and-support-section/special-needs-support/students-sharing/athena_a-big-fan-of-robotics/]

[<https://youtu.be/Oj3ewcWgY5c?si=zkdg8hLiU-TVv13f>]